

Vehicle Management System

Feature Guide - Barangay Emergency Response Vehicle Management

Group 4 | What the system does today

Overview

The Vehicle Management System (VMS) is a web-based desktop system that helps the barangay organize, monitor, and maintain its emergency response vehicles. It gives Admin, Custodian, and Maintenance Personnel a shared, accountable record of every vehicle - its condition, reported issues, repairs, schedules, and readiness to respond. This guide lists everything the system does today, including a recent round of enhancements (marked New) that make the system capture real-world situations - relayed reports, parts delays, repeat failures, emergency field repairs - that it previously had no place to record.

Platform & Access

- Web desktop application with role-based access for Admin, Custodian, and Maintenance Personnel; one account may hold several roles.
- Secure login (rate-limited), logout, password change, and active/inactive account enforcement.
- Cloud file and image uploads - issue photos, repair attachments, and vehicle photos.
- User account management: create accounts, activate and deactivate users.

Vehicle Records

- Full vehicle records with create, edit, archive/restore, and decommission.
- Acquisition cost on file (optional) so lifetime repair spend can be compared against what the vehicle is worth. (New)
- Vehicle categories/types and locations/hubs (with map coordinates).
- Vehicle statuses: Available, Under Maintenance, Inactive, Decommissioned.
- Ready-to-Respond readiness: readiness checks with 24-hour freshness, readiness states, a Ready-to-Respond column and stat cards, plus add/remove of readiness-check items.
- Reliability view per vehicle: failures in the last 6/12 months, average days out of service, lifetime repair spend, and a chronic-unit warning.
- Decommission signal: when lifetime repair spend reaches 60% or more of the vehicle's acquisition cost, the reliability view flags it as a decommission-review candidate. (New)

Condition Monitoring

- Periodic custodian condition checks recording each vehicle's physical condition over time.
- Create Ticket from a "Needs Repair"/"Damaged" condition check: the check already found the problem, so this spawns a pre-diagnosed ticket straight into repair assignment instead of re-inspecting what was just inspected. (New)

Issue Reports

- Issue report records with photo evidence, severity level, and status tracking.
- Duplicate / open-issue warning - every currently open issue already on a vehicle is shown as soon as one is picked, so two people relaying the same problem don't silently create two reports. (New)
- One-click Create Ticket directly from an issue report, carrying its category and severity in.
- "Reported on behalf of [driver]" - an optional field that keeps the real eyewitness on a relayed report, without

giving the driver a login (drivers are not system users). (New)

Maintenance Records

- Standalone maintenance records with a Verify (Custodian) then Confirm (Admin) lifecycle - usable without a pre-existing ticket.
- Progress states: Assigned, Under Repair, On Hold - Awaiting Parts, For Verification, Completed.
- "On Hold - Awaiting Parts" state distinguishes a parts delay from a slow mechanic, so the record doesn't just look stalled. (New)
- Maintenance Personnel can self-file a record for a field repair with no ticket yet; Admins are notified automatically; a new Issue Report can be created inline from the same form. (New)
- External-shop repairs: mark a record as external, capture the vendor name and a warranty-until date, so a repeat failure under warranty isn't paid for twice. (New)
- Field-level edit trail on sensitive edits (dates, cost, status, warranty) - the activity log records exactly what changed from what to what, not just that a record was touched. (New)

Maintenance Schedules (Preventive)

- Preventive schedules that, on completion, auto-generate the next recurrence at a chosen interval (including a custom number of months) while avoiding double-booking.
- Default "Scheduled only" view with pagination.
- Overdue / due-soon watch: any Scheduled preventive maintenance that's overdue or due within 7 days is surfaced on the dashboard as an at-risk signal instead of sitting silently on a calendar. (New)

Ticket Workflow (Request -> Validate -> Approval engine)

- Entry mode at creation - Inspection mode (today's default: Custodian diagnoses first) or Pre-Diagnosed mode, for when the problem is already known (relayed by phone, found during a condition check). A pre-diagnosed ticket is born Active with its sub-issues already listed, skipping a redundant re-inspection, and is stamped so the audit trail is honest about what was skipped and why. (New)
- Standardized fault category on every ticket (same catalog as issue types), so recurrence and cost tracking can group reliably instead of matching free-text titles. (New)
- Down-since timestamp, settable at creation (including backdated) so downtime figures reflect when the vehicle actually became unavailable, not just when the ticket was typed. (New)
- Standard flow: Create (Admin) -> assign Custodian -> Inspect (result Needs Maintenance or No Issues, creating sub-issues) -> Assign mechanic per sub-issue -> Log repairs (Mechanic) -> Verify (Custodian functional-test checklist + attestation) -> Confirm (Admin) -> Done.
- Add a sub-issue mid-ticket; reassign a mechanic; reopen/unconfirm a Done sub-issue; defer a sub-issue (opens a breadcrumb issue report).
- Close, cancel/uncancel, or delete a ticket; assignment-based verification (only the assigned custodian may verify).
- Ticket archive audit log with the ability to reopen an accidentally deleted ticket.
- "It broke again" recurrence link: a new ticket for the same fault (by category) within 90 days of a prior closed ticket is linked to it directly and flagged to Admins by name - "Nth time... last closed as fixed on Ticket #X" - not just a bare count. (New)

Dashboards & Oversight

- Role-specific dashboards for Admin, Custodian, and Maintenance.
- Emergency Readiness dashboard card with verified-ready as the headline number, per vehicle category.

- Fleet-level readiness summary: one headline figure - "X of Y emergency vehicles verified ready right now" - with a hard No Coverage alert the moment any vehicle category has zero ready units. (New)
- Reports (vehicle, issue, maintenance, history), activity logs, and a per-vehicle history timeline.
- Color-coded notifications by type, with read/unread tracking.
- Work Tracker - Custodian and Maintenance see the vehicles they have worked on and the outcomes, with filters.

Appendix - Consciously Out of Scope

These are real needs deliberately left out because they fall outside the project scope (web desktop only; roles limited to Admin/Custodian/Maintenance; no operations/dispatch, inventory, or mileage). They are noted so the boundary is a conscious choice, not an oversight.

- Offline / mobile field capture - scope is a web desktop platform.
- Dispatch-anyway override and post-emergency-use checks - require operations/deployment tracking.
- Consumables/stock readiness (oxygen, water, fuel) and parts cannibalization - require an inventory domain.
- Shift-roster handoff - requires an on-duty roster concept.

Of these, consumables readiness is the most emergency-relevant candidate should the scope ever be widened.